Number of Closed Islands (easy)

Problem Statement

You are given a 2D matrix containing only 1s (land) and 0s (water).

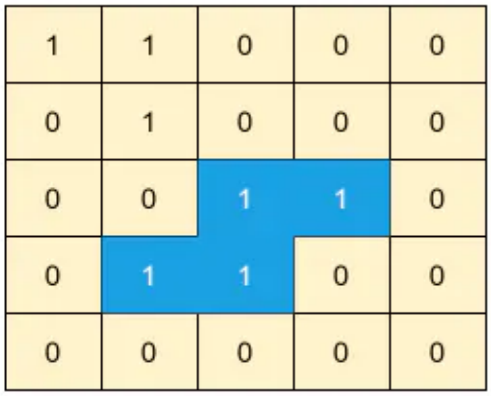
An **island** is a connected set of 1s (land) and is surrounded by either an edge or 0s (water). Each cell is considered connected to other cells horizontally or vertically (not diagonally).

A **closed island** is an island that is totally surrounded by 0s (i.e., water). This means all horizontally and vertically connected cells of a closed island are water. This also means that, by definition, a closed island can't touch an edge (as then the edge cells are not connected to any water cell).

Write a function to find the number of closed islands in the given matrix.

Example 1

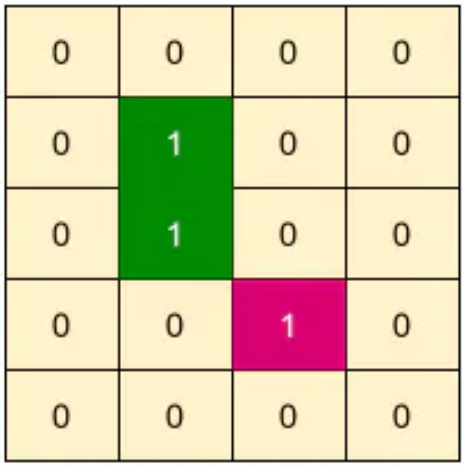
Input: matrix =



Output: 1 Explanation: The given matrix has two islands, but only the highlighted island is a closed island. The other island is touching the boundary that's why is is not considered a closed island.

Example 2

Input: matrix =



Output: 2

Explanation: The given matrix has two islands and both of them are closed islands.

Try it yourself

Try solving this question here:

Java

```java
import java.util.*;

class Solution {
public int countClosedIslands(int[][] matrix) {
    int countClosedIslands = 0;
    // TODO: Write your code here
    return countClosedIslands;
  }
}

```

Test Case | Run Code Result | Stdout

```
[[1,1,0,0,0],[0,1,0,0,0],[0,0,1,1,0],[0,1,1,0,0],[0,0,0,0,0]]
```

Use Example Testcases | Run | Submit